33. What is Kubernetes Scheduler?

Professional Answer

The Scheduler assigns pods to nodes based on resource availability, constraints, affinity rules, and policies.

Unprofessional Answer

Scheduler decides where pods run.

What is the role of kube-scheduler?

The kube-scheduler is responsible for assigning pods to appropriate worker nodes.
It watches for newly created pods that do not have a node assigned.
The scheduler evaluates nodes based on resource availability and scheduling constraints.
It considers factors like CPU, memory, taints, tolerations, and affinity rules.
After selection, it updates the pod specification with the chosen node name.

34. What factors does the scheduler consider?

Professional Answer

- CPU & memory requests
- Node taints & tolerations
- Pod affinity/anti-affinity
- Node selectors
- Topology constraints

Unprofessional Answer

Resources and rules.

3. kube-scheduler (Scheduling Internals)

What it does internally

Scheduler works in **two main phases**:

Phase 1: Filtering (Predicates)

Eliminates nodes that cannot run the pod:

- Insufficient CPU/memory
- Taints not tolerated
- NodeSelector mismatch

- Volume conflicts

Phase 2: Scoring (Priorities)

Scores remaining nodes:

- Least loaded node
- Pod affinity
- Topology spread

Final Step

- Highest score node selected
- Writes node name to PodSpec via API server

Important

Scheduler **never runs containers**, only assigns nodes

Nice 👍 — **kube-scheduler** is a *favorite* for **design + reasoning interviews** because it tests how you think, not just what you memorize.

Below is a **complete senior-level scheduler package**:

- Interview Q&A (basic → advanced)
- Scheduling flow (step-by-step)
- Troubleshooting & commands
- Real production scenarios
- Ultra-hard interview questions

Kubernetes kube-scheduler – Interview Q&A + Troubleshooting + Scenarios

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PART 1 Core Interview Questions

Q1. What is kube-scheduler?

✅ **Professional Answer**

kube-scheduler is a control plane component responsible for selecting a suitable worker node for a Pod.
It watches for Pods that do not have a node assigned.
It evaluates nodes based on resources and constraints.

It assigns the Pod to the best-fit node.
It does not run Pods itself.

❌ **Weak Answer**

Scheduler decides where pods go.

Q2. When does kube-scheduler act?

✅ **Professional Answer**

kube-scheduler acts only when a Pod is created without a node assignment.
It does not manage running Pods.
It does not reschedule Pods automatically.
Once a Pod is bound to a node, the scheduler is done.
Re-scheduling happens only after Pod recreation.

Q3. Does kube-scheduler run on worker nodes?

✅ **Professional Answer**

No, kube-scheduler runs on control plane nodes.
It runs as a static Pod or system service.
Multiple scheduler instances can run for HA.
Only one instance becomes leader at a time.
Leader election prevents double scheduling.

PART 2 Scheduling Flow (Very Important)

Q4. Explain the kube-scheduler decision flow.

✅ **Professional Answer**

The scheduler follows three major steps.
First, it filters nodes that do not meet Pod requirements.
Second, it scores the remaining nodes.
Third, it selects the highest-scoring node.
Finally, it binds the Pod to that node.

Interview line:

Filter → Score → Bind

Q5. What are scheduling constraints?

✅ **Professional Answer**

Scheduling constraints restrict where a Pod can run.
They include resource requests, taints, tolerations, and affinity rules.

Node selectors and topology constraints also apply.
If no node satisfies constraints, the Pod stays Pending.
Constraints provide placement control.

PART 3 Key Scheduling Concepts (High Interview Value)

Q6. What is nodeSelector?

✅ **Professional Answer**

nodeSelector is a simple key-value matching rule.
It restricts Pods to nodes with specific labels.
It is hard scheduling.
If no node matches, the Pod remains Pending.
It is easy but limited.

Q7. Node affinity vs nodeSelector?

✅ **Professional Answer**

Node affinity is more expressive than nodeSelector.
It supports required and preferred rules.
Preferred rules influence scoring but are not mandatory.
Required rules are strict like nodeSelector.
Affinity allows flexible scheduling.

Q8. What are taints and tolerations?

✅ **Professional Answer**

Taints repel Pods from nodes.
Tolerations allow Pods to run on tainted nodes.
They are used for node isolation.
Without toleration, Pods are not scheduled.
This protects special-purpose nodes.

Q9. What is Pod affinity and anti-affinity?

✅ **Professional Answer**

Pod affinity attracts Pods to specific Pods.
Pod anti-affinity separates Pods for high availability.
They work using Pod labels.
They support topology awareness.
They improve fault tolerance.

PART 4 Scheduler Troubleshooting (Step-by-Step)

Scenario 1 Pods stuck in Pending

Troubleshooting Steps

kubectl describe pod <pod>

Check:

- Insufficient CPU or memory
- Taints without tolerations
- Node affinity mismatch
- Volume binding issues

Scenario 2 Scheduler is Down

Symptoms

- New Pods never leave Pending
- Existing Pods unaffected

Steps

kubectl get pods -n kube-system | grep scheduler

journalctl -u kube-scheduler

Scenario 3 Pod scheduled on unexpected node

Investigation

- Check node affinity and tolerations
- Check preferred vs required rules
- Check resource availability

Scenario 4 Pod scheduling slow in large clusters

Root Causes

- Too many nodes
- Complex affinity rules
- Inefficient scheduler configuration

Scenario A

High-priority Pod not scheduled even when nodes are free

Expected Answer

- Priority class misconfiguration
- Resource requests too high
- Taints blocking scheduling

Scenario B

Pods evenly spread initially, then all land on one node

Expected Answer

- Pod anti-affinity not defined
- Default scheduler behavior
- Resource fragmentation

Scenario C

DaemonSet Pods schedule correctly but Deployment Pods don't

Expected Answer

- Taints tolerated by DaemonSet
- Deployment missing tolerations

Scenario D

Pods Pending with "volume node affinity conflict"

Expected Answer

- Persistent volume zone mismatch
- Stateful workload misplacement

PART 6 Ultra-Hard Scheduler Interview Questions

Q1 Does kube-scheduler reschedule Pods if a node fails?

Answer

No.
The scheduler does not move running Pods.

Controllers recreate Pods.
Scheduler only assigns new Pods.

Q2 How does scheduler handle preemption?

Answer

Scheduler may evict lower-priority Pods.
This frees resources for higher-priority Pods.
Preemption is controlled by PriorityClass.
It must be used carefully.

Q3 Can you run multiple schedulers?

Answer

Yes.
Custom schedulers can coexist.
Pods select scheduler using schedulerName.
Useful for specialized workloads.

Q4 Scheduler vs Controller Manager?

Answer

Scheduler places Pods.
Controller Manager ensures desired state.
Scheduler makes one-time decisions.
Controllers continuously reconcile.

PART 7 Golden Interview Lines 🏆

Scheduler decides placement once; controllers ensure it stays correct.

Pending Pods always mean scheduling constraints or scheduler issues.

Scheduler never moves running Pods — that's a controller's job.

PART 8 Quick Comparison Table

Component	Role
Scheduler	Pod placement
kubelet	Pod execution
kube-proxy	Service routing